

2D Flow Assumption and Vorticity Explanation

General 3D Velocity Field

$$\mathbf{V} = u(x,y,z)\mathbf{i} + v(x,y,z)\mathbf{j} + w(x,y,z)\mathbf{k}$$

A fluid particle can move in all three directions: x, y, and z.

2D Flow Assumption

For a true two-dimensional flow:

$$u = u(x,y), v = v(x,y), \text{ and } w = 0.$$

Therefore, $\mathbf{V} = u(x,y)\mathbf{i} + v(x,y)\mathbf{j}$.

The flow exists entirely in the x-y plane with no motion in the z-direction.

Vorticity Definition

$$\boldsymbol{\omega} = \nabla \times \mathbf{V}$$

For a general 3D flow:

$$\omega_x = \partial w / \partial y - \partial v / \partial z$$

$$\omega_y = \partial u / \partial z - \partial w / \partial x$$

$$\omega_z = \partial v / \partial x - \partial u / \partial y$$

Applying the 2D Assumptions

Since $w = 0$ and there is no dependence on z, all z-derivatives vanish.

Thus:

$$\omega_x = 0$$

$$\omega_y = 0$$

$$\omega_z = \partial v / \partial x - \partial u / \partial y$$

Result

Only the z-component of vorticity survives:

$$\boldsymbol{\omega} = (\partial v / \partial x - \partial u / \partial y) \mathbf{k}$$

The velocity vectors lie in the x-y plane, so any fluid rotation occurs about an axis perpendicular to that plane (the z-axis).

Why $\omega_z = \boldsymbol{\omega} \cdot \mathbf{k}$?

The vorticity vector can be written as:

$$\boldsymbol{\omega} = \omega_x \mathbf{i} + \omega_y \mathbf{j} + \omega_z \mathbf{k}.$$

Taking the dot product with $\mathbf{k}$ extracts the z-component:

$$\boldsymbol{\omega} \cdot \mathbf{k} = \omega_z.$$

Importance in CFD

For 2D incompressible flow, only one vorticity value must be stored at each grid point. This simplifies the Navier-Stokes equations and leads to the streamfunction-vorticity formulation used in lid-driven cavity and other finite-difference solvers.